Orecast

Mineral intelligence over time

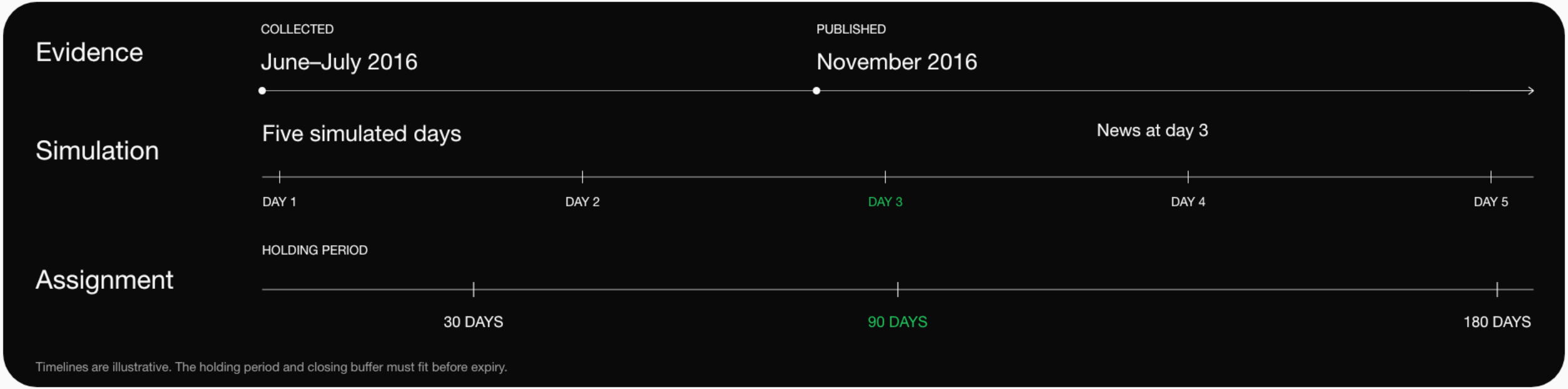


One hypothetical contract.
One fixed exposure, compared over time.

Assumed exposure: 23,076.92 troy oz and $184,615.38 buyer consideration.
Frozen invented-reference scale: 30,000 / 650. One contract, not multiple lots.

Historical signing payments do not value this invented mineral interest.

Fixed $50 entry / exit fees: 12.35% return. Proportional fees: 1.09% and a failed risk cap.



Timelines are illustrative. The holding period and closing buffer must fit before expiry.

We begin with

67

measured sediment samples from Tok River, Alaska.

A has 33 samples. B has 34.
Primary gold assays screen the evidence.

Recoverable ounces remain a separate assumption.

The evidence explorer

Orecast

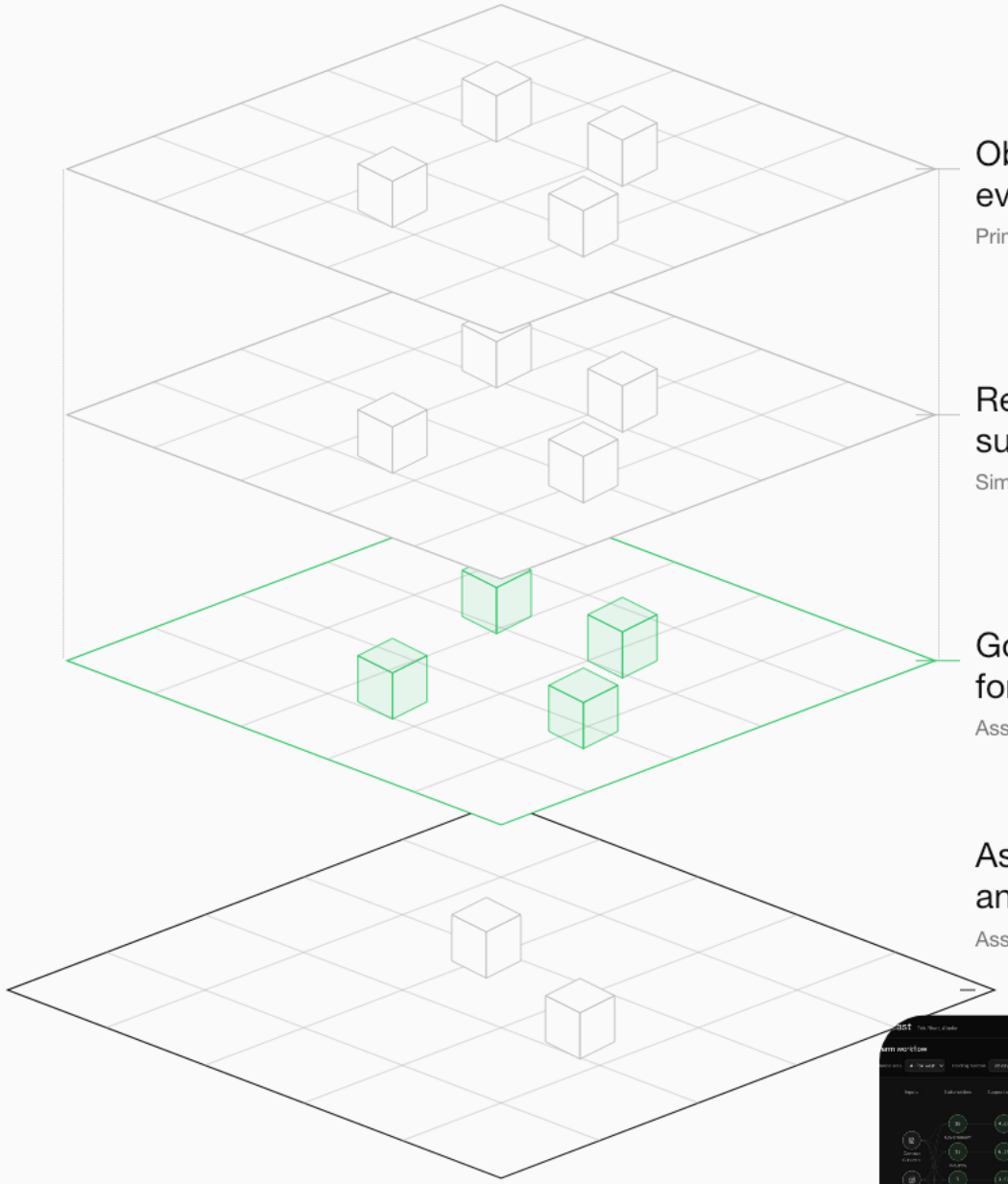


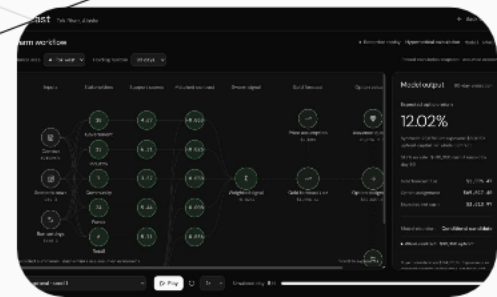
Alaska DGGS RDF 2016-10. Comparison areas, not available listings.



Orecast adjusts the gold forecast, then recomputes mineral and interest value, assignment and cash.

Observed assays screen evidence.
The synthetic exposure stays fixed.
Gold response, fees and liquidity are assumed.





Recorded support changes assumed gold. The same synthetic contract is held fixed.

Mine approval

–0.0252 support · $1,996.47/oz gold

$3,610.97 cash · 12.02% return

Court blockage

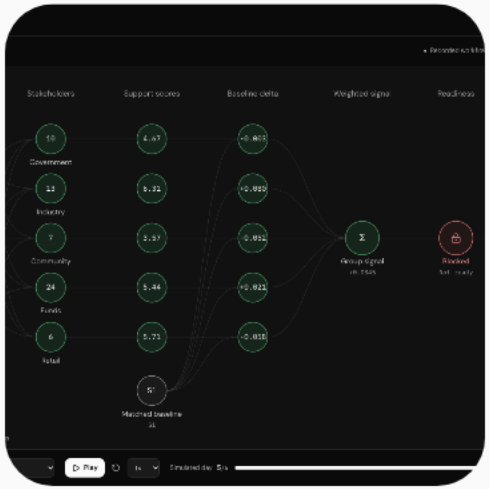
+0.0307 support · $2,004.30/oz gold

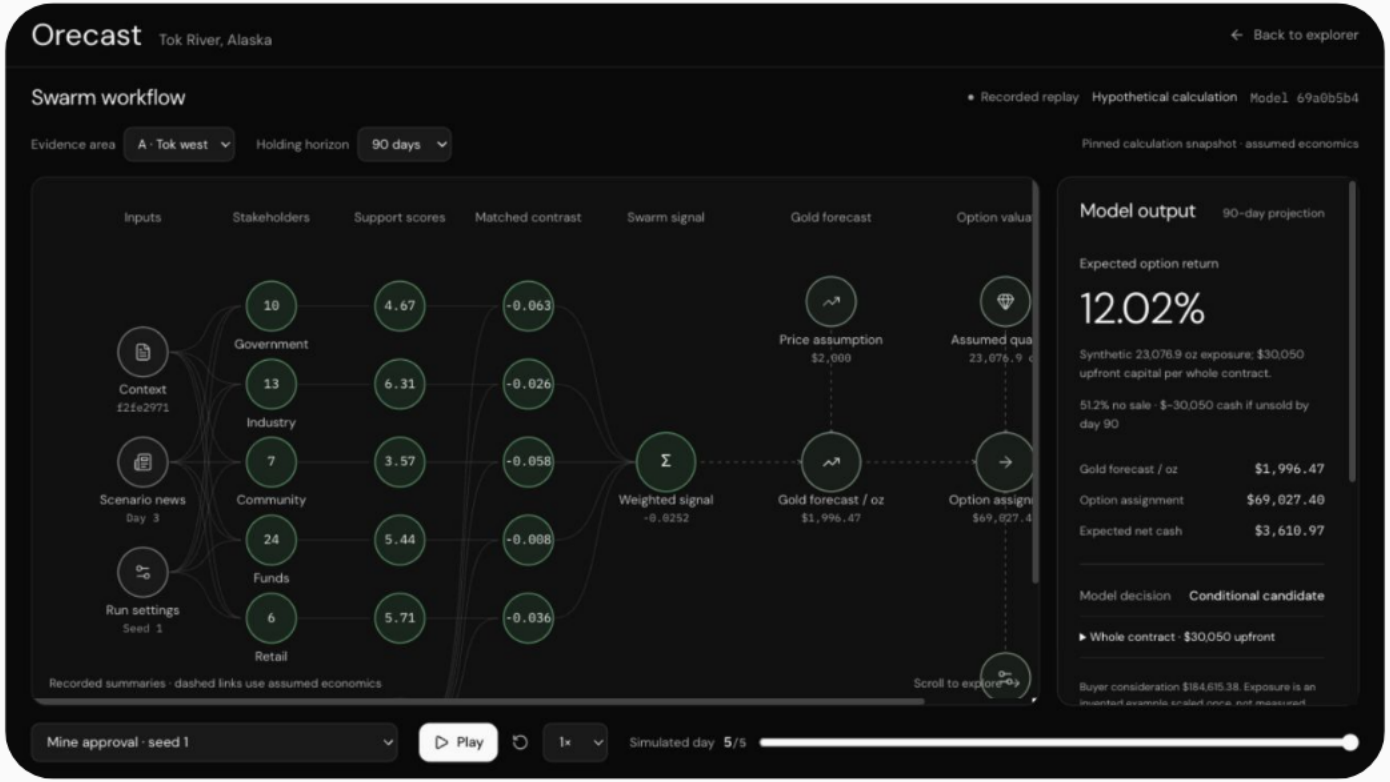
$3,831.36 cash · 12.75% return

Fixed $50 entry / exit fees.
Conditional candidates, not trades.
Research checks failed.
Gold response uncalibrated.



Matching sports-placebo reference, days 3–5.
Support scale –1 to +1. Area A, 90 days.





At $30,050 upfront,
the 90-day fixed-fee case has

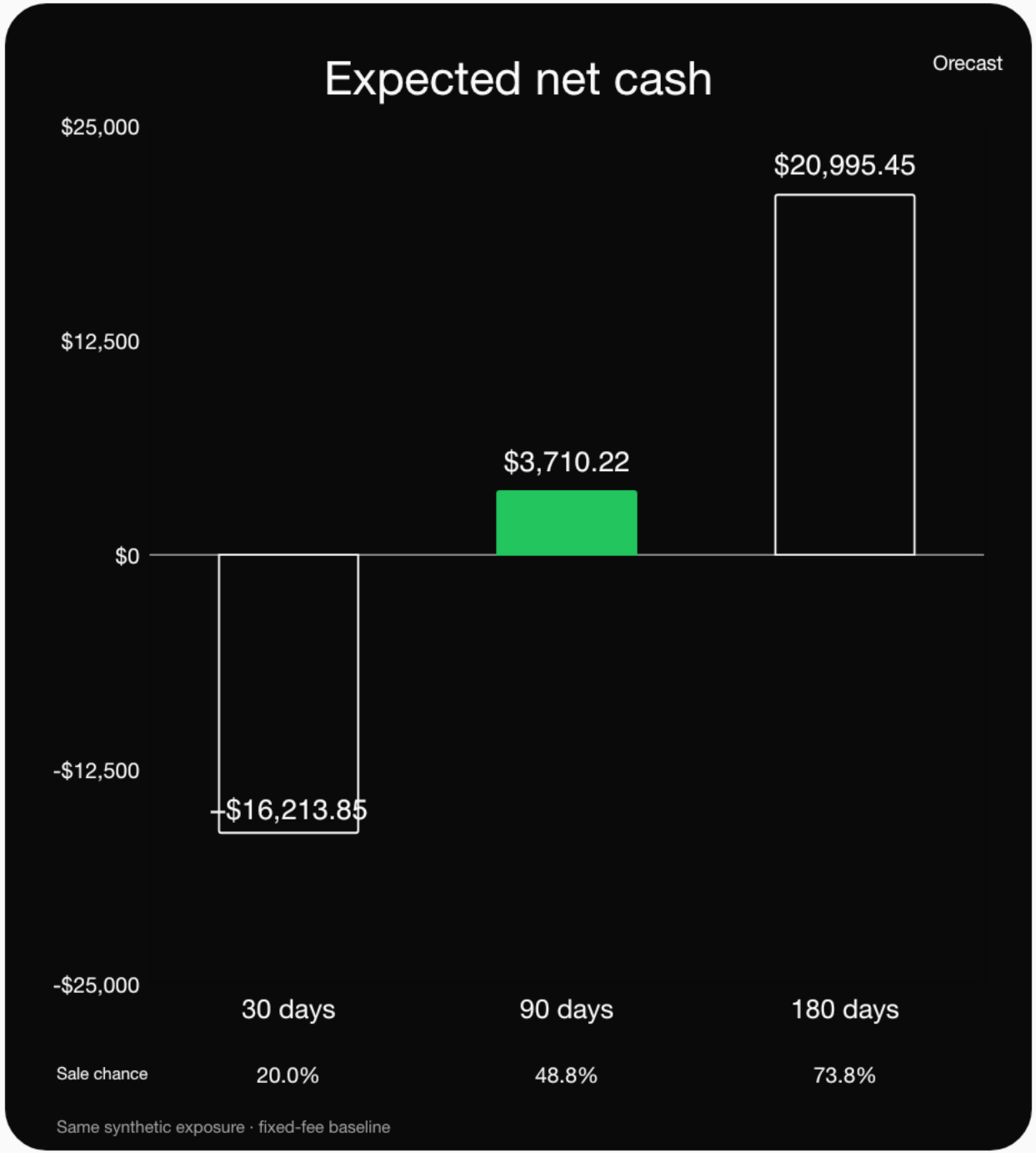
12.35%

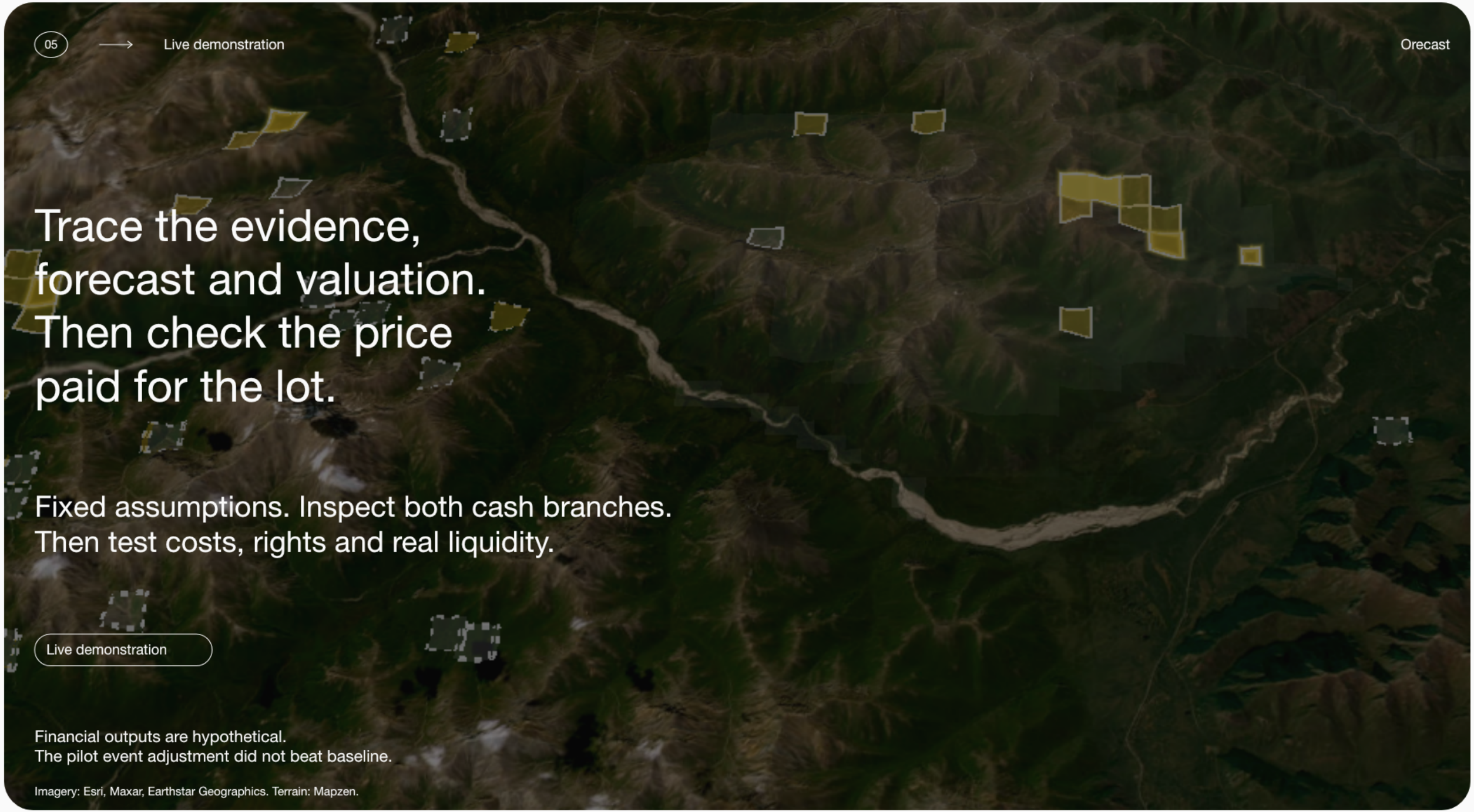
expected option return, while

51.2%

of paths have no sale.

Fixed $50 entry / exit fees. Median cash: –$30,050.
Proportional fees: 1.09% return; risk cap fails. See appendix 10.
Non-annualized. Gold growth zero; 20% assumed 30-day hazard.





Trace the evidence, forecast and valuation. Then check the price paid for the lot.

Fixed assumptions. Inspect both cash branches.
Then test costs, rights and real liquidity.

Live demonstration

Financial outputs are hypothetical.
The pilot event adjustment did not beat baseline.

Imagery: Esri, Maxar, Earthstar Geographics. Terrain: Mapzen.

Baseline cash and option return at 90 days

Hypothetical cash branches after all entry and exit costs. All amounts USD.

Outcome by day 90	Probability	Net cash	Weighted cash
Sale	48.8%	$39,130.77	$19,095.82
No sale	51.2%	–$30,050.00	–$15,385.60
Expected net cash	**100.0%**		**$3,710.22**

$$0.488 \times (\$69{,}230.77 - \$50) - \$30{,}050 \approx \$3{,}710.22$$

Expected option return: 12.35%, non-annualized. Conditional candidate.

For a 25% expected return, required conditional resale is $77,022.34.
A hurdle, not a forecast. Synthetic assignment is $69,230.77; no input was chosen from the target.

Historical pilot: the event adjustment did not beat baseline

12 selected equity events. Five held-out observations per horizon, in distinct cohorts. Lower MAE is better.

Model	5-session MAE (pp)	20-session MAE (pp)
No change	2.5562	5.9474
Gold-only	3.1291	7.5670
Gold + event	8.0596	14.3108

Retrospective equity component test, not Tok option returns.

Event adjustment did not beat baseline.
No paired historical swarm outcomes.

Next: paired prospective outcomes and calibrated assignment liquidity.

The fee assumption changes the result

One synthetic contract, the same gold forecast and sale hazard. Returns over 90 days are non-annualized.

Fee assumption	Entry fee	Exit fee	Upfront capital	Expected return
Fixed fees	$50.00	$50.00	$30,050.00	12.35%
Proportional fees	$2,307.69	$2,307.69	$32,307.69	1.09%

Proportional fee = $50 × (30,000 / 650) = $2,307.69 per fee.

$350.77 expected cash / $32,307.69 = 1.09% return.

$32,307.69 exceeds the $30,050 per-option cap. The model skips this case.

The higher fixed-fee result is an assumed sensitivity, not a measured margin.